



Eric Araújo

Associate Professor

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Welcome

I investigate the spread of opinions, sentiments, and behaviors in complex systems through multiagent modeling and simulations, offering insights into social dynamics in politics, cooperation, health, criminology, and religion.

The work I develop is multidisciplinary, using agent-based models and data science to solve problems in areas such as political polarization in churches, public security, the spread of fake news, and promoting healthier lifestyles.

Education

Ph.D. Vrije Universiteit Amsterdam , Computer Science • Thesis: Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks	Amsterdam, NL Aug 2014 – Sept 2018
M.Sc. Universidade Federal de Minas Gerais , Computer Science	Belo Horizonte, Brazil Mar 2007 – Sept 2009
B.Sc. Universidade Federal de Viçosa , Computer Science	Viçosa, Brazil Mar 2003 – Dec 2006

Experience

Calvin University , Associate Professor • Computer Science Department Calvin University	Grand Rapids MI USA July 2024 – present 2 years 3 months
FIOCRUZ , Researcher • FIOCRUZ is a leading institution in health research and development in Brazil. My work focused on collecting data and analyzing it to understand medical circulation of health sciences among Brazilian researchers.	Rio de Janeiro Brazil Jan 2022 – Jan 2024 2 years 1 month
Vrije Universiteit Amsterdam , Guest Researcher • Collaborator in the Department of Computer Science Vrije Universiteit Amsterdam • Worked on criminology models with Prof. Dr. Charlotte Gerritsen.	Oct 2018 – Apr 2022 3 years 7 months
Universidade Federal de Lavras , Associate Professor • Computer Science Department Universidade Federal de Lavras	Minas Gerais Brazil Feb 2011 – June 2024 13 years 5 months
Gammon Presbyterian Faculty , Professor • Professor and Coordinator of the Information Systems course	Minas Gerais Brazil Jan 2010 – Jan 2011 1 year 1 month

Books

Revisiting AI Ethics: A Practice-Grounded Approach for Development, Implementation and Policy Making

Forthcoming 2026

Eric Araújo, Maarten Verkerk, Christine Boshuijzen-van Burken

(Springer. Under contract; manuscript due August 2026. Provisional title. Applies Dooyeweerd's modal framework to the ethics of artificial intelligence.)

Publications

Context-dependent leadership in global health science: Brazil's role across North-South, South-South, and triangular collaboration models (2010–2019)

Aug 2026

Juliana Freitas Lopes, Priscila Costa Albuquerque, **Eric Araújo**, Fabio Zicker, Bruna de Paula Fonseca

[10.1007/s11192-026-05781-0](https://doi.org/10.1007/s11192-026-05781-0) (Scientometrics)

Caracterização Temporal de Roubos de Rua Contrastando Contextos Pré-Pandêmico e Pandêmico

June 2026

Renato Figueiredo Frade, **Eric Fernandes de Mello Araújo**, João Paulo Roquim Romanelli, Carlos Henrique da Silveira

[10.31060/rbsp.2026.v20.n2.2081](https://doi.org/10.31060/rbsp.2026.v20.n2.2081) (Revista Brasileira de Segurança Pública)

From brain drain to brain circulation: International researcher mobility in Brazilian health and life sciences (2005–2020)

Feb 2026

Leonardo Biazoli, Priscila Costa Albuquerque, **Eric Araújo**, Izabela Regina Cardoso de Oliveira, Bruna de Paula Fonseca

[10.1162/QSS.a.411](https://doi.org/10.1162/QSS.a.411) (Quantitative Science Studies)

Exploring the link between urban topology and street crime using complex networks: a case study from Southeast Brazil

July 2025

Matheus de Andrade Flausino, **Eric Araújo**, Angélica Sousa da Mata

[10.1093/comnet/cnaf016](https://doi.org/10.1093/comnet/cnaf016) (Journal of Complex Networks)

The Use of Agent-based Modeling in the Study of Complex Systems (In Portuguese)

Jan 2025

Clayton R. da Silva, Olivia Mesquita, **Eric Araújo**, Angélica S. Mata

[10.1590/1806-9126-RBEF-2024-0464](https://doi.org/10.1590/1806-9126-RBEF-2024-0464) (Revista Brasileira de Ensino de Física)

Understanding the complexities of Bluetooth for representing real-life social networks

Jan 2024

Bojan Simoski, Michel C.A. Klein, **Eric Fernandes de Mello Araújo**, Aart T. van Halteren, Thabo J. van Woudenberg, Kirsten E. Bevelander, Moniek Buijzen, Henri Bal

[10.1007/s00779-020-01435-x](https://doi.org/10.1007/s00779-020-01435-x) (Personal and Ubiquitous Computing)

Police Planning Based on the Use of Business Intelligence (BI): The Police Manager's Perception Regarding the Violent Crime Rate in the 8th Battalion of the Military Police of Minas Gerais (PMMG) (In Portuguese)

Jan 2023

Erich da Costa, Denis Renato de Oliveira, **Eric Araújo**

[10.29377/rebsp.v16i01.614](https://doi.org/10.29377/rebsp.v16i01.614) (Revista Brasileira de Estudos de Segurança Pública)

Epidemiological Intelligence in Operation COVID-19: A New Decision Support Tool in Brazil's Military Logistics (In Portuguese)

Jan 2023

José Roberto Pinho de Andrade Lima, Mariza Ferro, **Eric Araújo**, Cristiano Barros, Ernesto Rademaker Martins, Beatriz Helena Felício Fuck Telles Ferreira

[10.47240/revistadaesg.v38i82.1250](https://doi.org/10.47240/revistadaesg.v38i82.1250) (Revista da Escola Superior de Guerra)

Political Polarization on Twitter During the COVID-19 Pandemic: A Case Study in Brazil Pedro Brum, Matheus Cândido Teixeira, Renato Vimieiro, Eric Araújo , Wagner Meira Jr, Gisele Lobo Pappa 10.1007/s13278-022-00949-x (Social Network Analysis and Mining)	Jan 2022
Urban Afforestation and Public Safety: A Bibliometric Study Using the Consolidated Meta-Analytical Focus Theory (In Portuguese) Kelly Iapueque Rodrigues de Sousa, Michele Valquíria dos Reis, Rafael Rodrigues de Castro, Eric Araújo 10.5585/geas.v11i2.22965 (Revista de Gestão Ambiental e Sustentabilidade (GeAS))	Jan 2022
Detecção e Classificação de Bots Utilizando Redes Neurais Artificiais e Análise de Sentimentos Gabrieli Silva, Eliaquim Ramos, Eric Araújo , Fábio Borges, Mariza Ferro www.eamc.lncc.br/proceedings.html (Proceedings of the XIV Encontro Acadêmico de Modelagem Computacional)	Jan 2021
The Virus and Socioeconomic Inequality: An Agent-based Model to Simulate and Assess the Impact of Interventions to Reduce the Spread of COVID-19 in Rio de Janeiro, Brazil Vinícius Prata Klôh, Gabrieli Dutra Silva, Mariza Ferro, Eric Araújo , Cristiano Barros de Melo, José Roberto Pinho de Andrade Lima, Ernesto Rademaker Martins 10.34119/bjhrv3n2-192 (Brazilian Journal of Health Review)	Jan 2020
Disconnecting for the Good: A Network-Oriented Model for Social Contagion of Opinions and Social Network Interventions to Increase Adherence to Social Distancing Eric Araújo , Mariza Ferro, Gabrieli Silva 10.5753/brasnam.2020.11170 (Proceedings of the IX Brazilian Workshop on Social Network Analysis and Mining)	Jan 2020
Creating a Temporal Pattern for Street Robberies Using ABM and Data from a Small City in Southeast Brazil Eric Araújo , Charlotte Gerritsen 10.4324/9780429277177 (Agent-Based Modelling for Criminological Theory Testing and Development)	Jan 2020
Identifying Influence Agents That Promote Physical Activity Through the Simulation of Social Network Interventions: Agent-Based Modeling Study Thabo J. van Woudenberg, Bojan Simoski, Eric Araújo , Kirsten E. Bevelander, William J. Burk, Crystal R. Smit, Laura Buijs, Michel Klein, Moniek Buijzen 10.2196/12914 (Journal of Medical Internet Research)	Jan 2019
A Social Network Model for Integration of Refugees Fabio Curi, Dimitris Nikolopoulos, Eric Araújo 10.5220/0007930601650175 (Proceedings of the 9th International Conference on Simulation and Modeling Methodologies, Technologies and Applications)	Jan 2019
Parameter Optimization for Deriving Bluetooth-Based Social Network Graphs Bojan Simoski, Michel C.A. Klein, Eric Araújo , Aart T. Van Halteren, Thabo Van Woudenberg, Kirsten E. Bevelander, Moniek Buijzen, Henri Bal 10.1109/SmartWorld-UIC-ATC-SCALCOM-IOP-SCI.2019.00318 (2019 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI))	Jan 2019

Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks (PhD Thesis) Eric Araújo research.vu.nl/en/publications/632878cc-e2e7-40ab-a34e-91bde40eaf50 (Vrije Universiteit Amsterdam)	Jan 2018
Using Simulations for Exploring Interventions in Social Networks - Modeling Physical Activity Behaviour in Dutch School Classes Eric Araújo , Bojan Simoski, Thabo van Woudenberg, Kirsten Bevelander, Crystal Smit, Laura Buijs, Michel Klein, Moniek Buijzen 10.5220/0006857704140425 (Proceedings of 8th International Conference on Simulation and Modeling Methodologies, Technologies and Applications - SIMULTECH)	Jan 2018
Applying Machine Learning Algorithms for Deriving Personality Traits in Social Network Eric Araújo , Bojan Simoski, Michel Klein 10.1145/3167132.3167377 (Proceedings of the 33rd Annual ACM Symposium on Applied Computing)	Jan 2018
Detecting Dutch Political Tweets: A Classifier based on Voting System using Supervised Learning Eric Araújo , Dave Ebbelaar 10.5220/0006592004620469 (Proceedings of the 10th International Conference on Agents and Artificial Intelligence - Volume 1: ICAART)	Jan 2018
Social Connection Dynamics in a Health Promotion Network Eric Araújo , Michel Klein, Aart van Halteren 10.1007/978-3-319-50901-3_61 (Complex Networks & Their Applications V)	Jan 2017
Explaining Changes in Physical Activity Through a Computational Model of Social Contagion Julia S. Mollee, Eric Araújo , Adnan Manzoor, Aart T. van Halteren, Michel Klein 10.1007/978-3-319-54241-6_19 (Complex Networks VIII)	Jan 2017
Exploring Parameter Tuning for Analysis and Optimization of a Computational Model Julia S. Mollee, Eric Araújo , Michel C. A. Klein 10.1007/978-3-319-60045-1_36 (Advances in Artificial Intelligence: From Theory to Practice)	Jan 2017
A Temporal-Causal Model for Spread of Messages in Disasters Eric Araújo , Annelore Franke, Rukshar Wagid Hosain 10.1007/978-3-319-67077-5_37 (Computational Collective Intelligence)	Jan 2017
A Computational Cognitive Model for Political Positioning and Reactions in Web Media Eric Araújo , Michel Klein 10.1109/ICCI-CC.2017.8109782 (2017 IEEE 16th International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC))	Jan 2017
Analysis and Refinement of a Temporal-Causal Network Model for Absorption of Emotions Eric Araújo , Jan Treur 10.1007/978-3-319-45243-2_3 (Computational Collective Intelligence)	Jan 2016

Online Sharing of Physical Activity: Does It Accelerate the Impact of a Health Promotion Program?	Jan 2016
Adnan Manzoor, Julia S. Mollee, Eric Araújo , Aart T. van Halteren, Michel C. A. Klein 10.1109/BDCloud-SocialCom-SustainCom.2016.40 (2016 IEEE International Conferences on Big Data and Cloud Computing (BDCloud), Social Computing and Networking (SocialCom), Sustainable Computing and Communications (SustainCom) (BDCloud-SocialCom-Sustain-Com))	
Analysis and Evaluation of Social Contagion of Physical Activity in a Group of Young Adults	Jan 2015
Eric Araújo , Anita V. T. T. Tran, Julia S. Mollee, Michel C. A. Klein 10.1145/2818869.2818922 (Proceedings of the ASE BigData & SocialInformatics 2015)	
An Analysis of TCP Protocol Issues in Asymmetric Networks and Existing Solutions (In Portuguese) - M.Sc. Thesis	Jan 2009
Eric Araújo hdl.handle.net/1843/BUBD-9JWQ36 (Universidade Federal de Minas Gerais)	
AnimEasy: Free Tool for Digital Inclusion Focused on Multimedia Using Inkscape (In Portuguese) - B.Sc. Senior Project	Jan 2007
Eric Araújo , Ricardo dos Santos Ferreira (VII Simpósio de Informática do Planalto Médio (SIPM 2007))	

Events

CompCOVID 2020	Brazil (online)
Panel on computing applications during the COVID-19 pandemic. In Portuguese.	
	June 2020
IX Brazilian Workshop on Social Network Analysis and Mining (BraSNAM 2020)	Brazil (online)
Conference presentation of BraSNAM 2020 paper (doi:10.5753/brasnam.2020.11170).	
	Dec 2020
Saúde Única — UFLA	Lavras, Brazil (online)
Talk on opinion contagion in social networks and its epidemiological impact. In Portuguese.	
	Feb 2021
ABC²-Tec 5ª Reunião — Associação Brasileira de Cristãos na Ciência	Brazil (online)
Talk on social contagion models at the Brazilian Association of Christians in Science. In Portuguese.	
	July 2021
KDMiLe 2021 — Symposium on Knowledge Discovery, Mining and Learning	Brazil (online)
Keynote session (KSN2) at KDMiLe 2021. In Portuguese.	
	Oct 2021
PROCORES Research Group	Brazil (online)
Research group talk on social network spreading models. In Portuguese.	
	Dec 2022
Oficina Prograd — UFLA	Lavras, Brazil
Workshop on AI tools for undergraduate education. In Portuguese.	
	Mar 2024
Departamento de Estatística — UFLA	Lavras, Brazil
Invited talk on data-driven agent-based modeling. In Portuguese.	
	Oct 2024
19º Congresso Brasileiro de Sistemas (CBS)	Goiânia, Brazil
In Portuguese	
	Nov 2024
Autonomous Agents for Social Good (AASG) 2025	Detroit, MI
Presented by students Daniel Kwon and Jaden Brookens (senior project)	
	May 2025

Algoritmos e Sociedade — UFG In Portuguese	Goiânia, Brazil May 2025
INSNA Sunbelt 2025 With Juliana Freitas Lopes et al. (Fiocruz)	Paris, France June 2025
American Scientific Affiliation (ASA) Annual Meeting 2025	Colorado Christian University, Lakewood, CO July 2025
Grace Hopper Celebration 2025 The Grace Hopper Celebration is the world's largest gathering of women in computing. Attended with the Calvin Girls Who Code student delegation.	Chicago IL USA Nov 2025
Baylor Symposium on Faith & Culture 2026 — Technology and the Human Person in the Age of AI	Waco, TX Feb 2026
NetLogo Conference 2026 Presented by students Katelin Jandris, Ryan Klein, and Ovgu Tufan (senior project)	Chicago, IL July 2026
Prompted: Liberal Arts in the Age of AI — Season 2, Episode 6 Podcast guest, discussing the Matristics summer research on what leading large language models know about two fourth-century Christian women, and previewing the October 8–10 "Wisdom in the Age of AI" conference. Hosted by Craig Mattson; production funded by the Calvin Center for Christian Scholarship.	Calvin University (podcast) Aug 2026
American Academy of Religion (AAR) Annual Meeting 2026 — History of Christianity Unit Papers session "History of the Present, the Futures of Christianity's Past" (A23-217), Monday Nov 23, 12:30–2:30 PM. Empirical evaluation of LLM accuracy on Macrina the Younger and Olympias of Constantinople.	Denver, CO Nov 2026

Grants and Fellowships

Templeton World Charity Foundation — Planning Grant \$100,000 planning grant for a research program on polarization and listening across communities, awarded under the Templeton World Charity Foundation's Listening and Learning in a Polarized World (LLPW) program and co-led with Dr. Damaris Parsitau. Extends the BBB (Belief, Behavior, Belonging) agent-based modeling work to congregations in Brazil and Kenya, with ethnographic fieldwork preceding the computational modeling. Full implementation proposal due January 15, 2027.	June 2026 – Jan 2027
Marjorie G. Hoozeboom Student Faculty Research Fellowship Inaugural award of the Hoozeboom fellowship, supporting a 10-week summer research placement for student David Quinn on the empirical validation track of the TWCF church-polarization project: survey-data pipelines for initializing agent belief distributions, calibration targets drawn from the social science literature, and support for the international pilot survey instruments.	Summer 2026
Calvin Research Fellowship Awarded 4 faculty load hours for the 2026–2027 academic year for the project "Modeling Political and Religious Polarization in Christian Communities." Named a Diekema Fellow, with support paid through the Diekema Endowment.	2026 – 2027
Nagel Institute Fellowship Supporting research on computational modeling of world Christian communities.	Jan 2026

McGregor Summer Research Fellowship
Supporting collaborative faculty-student research.

May 2026 – Aug 2026

Teaching

Introduction to Data Structures

An introduction to data structures and algorithms, focusing on the design and analysis of algorithms, data structures, and their applications.

Calvin University USA
Sept 2024 – present

Databases Management Systems

An introduction to database management systems, covering data modeling, query languages, and database design.

Calvin University USA
Jan 2025 – present

Agent-based Modeling

An introduction to agent-based modeling, focusing on the design and implementation of agent-based models for complex systems.

Calvin University USA
Sept 2024 – present

Concepts of Programming Languages

Concepts and paradigms of programming languages.

Calvin University USA
Jan 2026 – present

Research Projects

Matristics: Evaluating Large Language Models on the Women of the Early Church

With D. Rylaarsdam. Supported by a McGregor Undergraduate Research Fellowship.

- A public benchmark testing whether leading large language models (Claude, ChatGPT, Gemini, Grok) reproduce the androcentric silences of the historical record about women in early Christianity.
- First paper accepted to the American Academy of Religion Annual Meeting 2026.

Calvin University
2026 – present

CS Prayers — A Book of Hours for Computing

Founder and curator.

- A public collection of liturgies for the rhythms of studying, teaching, and practicing computer science, open to contributions from students and faculty at any institution.

Calvin University
2026 – present

Procores

Research group funded by CNPq (Brazil)

- ProCoReS is a research group funded by CNPq, dedicated to the study of social networks, mathematical modeling, contagion processes, and the analysis of complex data. The group brings together researchers from different fields to propose and analyze innovative models that consider multiple aspects of social networks and their dynamics.
- To model networks and contagion, most current approaches use graphs. However, these are not entirely realistic, and new models of contagion processes should allow for: representation of different classes of individuals and connections, linkage between concurrent and co-occurring processes, and connections involving two or more individuals simultaneously. Understanding how connections are formed and how different types of flows permeate such networks is crucial to understanding how our society is organized and evolves. The goal is to propose mathematical, statistical, agent-based, and machine learning models that account for: multiple complex aspects of social contagion processes and parameterization through data sources (e.g., online social networks, mobility, collaborations). Such models should allow for classes of individuals and relationships; model co-occurrence, causality, and links between contagion processes; and represent multi-relationships.
- Funded by: CNPq Universal 2022

UFMG Brazil
May 2022 – present

Socioeconomic Impacts in Territories Affected by Technological Disasters

UFLA Brazil
Sept 2022 – Feb 2024

- This project investigates the socioeconomic consequences of technological disasters in Brazil, which have increased significantly in recent decades. Between 2000 and 2020, Brazil recorded 96 technological disasters—ranging from dam collapses to oil spills—impacting over 42 million people and causing over USD 17 billion in damages (EM-DAT, 2021). Unlike natural disasters, which occur without human interference, technological disasters result from anthropogenic actions, such as industrial accidents or environmental contamination. These events often lead to material loss, environmental degradation, and serious public health issues, demanding robust mechanisms for community response and risk management.
- By analyzing historical data and specific case studies such as the Mariana (2015) and Brumadinho (2019) dam collapses, the project seeks to understand how these disasters affect the social and economic fabric of impacted territories. The ultimate goal is to identify patterns and develop frameworks that can inform policies for disaster prevention, mitigation, and recovery—contributing to more resilient communities in the face of growing technological risks.

Analysis of User Behavior on a University Campus Wi-Fi Network

UFLA Brazil
Mar 2021 – Sept 2023

Masters Project by Thiago do Prado Ramos

- This project aims to analyze the behavior of users on a university campus Wi-Fi network, focusing on connection patterns and user mobility. By examining connection logs, the project seeks to understand how users interact with the network, their mobility patterns, and how these factors influence network performance and user experience.
- The findings will help in optimizing network infrastructure, improving user experience, and informing future network design and management strategies.

Algorithms for the Optimization of Supply Chain Management Systems

UFLA Brazil
Mar 2021 – Mar 2022

Project in partnership with Prof. Mayron Moreira (UFLA Brazil)

- This project developed and evaluated algorithms aimed at optimizing key components of supply chain management, including inventory control, logistics, and production scheduling. By modeling supply chains as dynamic systems, the project applied heuristic and metaheuristic approaches—such as genetic algorithms and simulated annealing—to improve efficiency, reduce costs, and enhance decision-making under uncertainty.
- The research focused on real-world constraints such as demand variability, delivery time windows, and resource limitations. The resulting algorithms demonstrated significant improvements in performance when compared to traditional methods, and provided a flexible framework that can be adapted to various industrial sectors seeking to enhance their operational processes.

Urban Tree Cover and Crime

Project in partnership with Prof. Michele Valquíria dos Reis (UFLA Brazil)

UFLA Brazil
Mar 2020 – Mar 2024

- Urban tree cover provides vital ecosystem services that promote public health and well-being. However, there is a common perception—found in media reports and informal discussions—that trees may facilitate crime by obstructing visibility or serving as hiding places. While some studies have linked vegetation to fear of crime, often recommending vegetation removal to improve safety, research from the Northern Hemisphere increasingly suggests a negative correlation between urban greenery and crime rates, indicating that vegetation may actually contribute to crime reduction.
- This project aims to investigate the relationship between urban tree cover and crime incidence in Brazilian cities. It integrates data from street tree inventories, georeferenced crime records from the Military Police, socioeconomic indicators from the national census (IBGE), citizen surveys on perceptions of vegetation and safety, and citizen requests for tree removal. The goal is to generate scientific insights into the social dimensions of urban greenery and to inform evidence-based environmental management policies. The findings are intended to support public agencies and civil society in designing safer, greener urban spaces.

Computational Modeling and Simulation of Contagion in Social Networks

UFLA Brazil
Sept 2018 – June 2024

- Social contagion refers to the spread of behaviors, perceptions, and emotions through interactions among individuals. Often occurring unconsciously, this phenomenon plays a role in shaping collective opinion, promoting positive behaviors, and influencing public discourse. This project aims to simulate the diffusion of information originating from social media platforms across various contexts, using agent-based and network modeling approaches.
- The study investigates how network topology, individual agent characteristics, and intervention strategies affect the spread and quality of information. It explores methods for enhancing the visibility and influence of high-quality content while limiting the reach of misinformation or fake news. The results aim to inform the design of healthier and more reliable information ecosystems, with potential applications in communication policy, platform design, and digital education.

Supervised Students

David Quinn

B.S. - Empirical Validation and Data Pipelines for an Agent-Based Model of Church Polarization

Calvin University
Summer 2026

- Marjorie G. Hoozeboom Student Faculty Research Fellowship

Miriam Crisman

B.S. - NetLogo Agent-Based Model of Church Community Polarization

Calvin University
Summer 2026

- Funded by the Templeton World Charity Foundation planning grant

Brady Swiech, Tim David, Zach Redder

B.S. - Scale66 — AI Platform for Marketing Content Generation

Calvin University
Fall 2025 - Spring 2026

Esther Asuquo, David Barry

B.S. - Modeling Gram-Negative Bacterial Infections: An Agent-Based Approach

Calvin University
Spring 2025 - Fall 2025

Katelin Jandris, Ryan Klein, Ovgu Tufan

B.S. - Cognitive Modeling of Church Polarization

Calvin University
Fall 2025 - Spring 2026

- Presented at NetLogo Conference 2026 (Chicago — July 2026)

Jaden Brookens, Daniel Kwon

B.S. - Modeling Introversion in the Classroom: An Agent-Based Approach

Calvin University
Fall 2024 - Spring 2025

- Conference paper at AASG 2025 (Detroit)

Leonardo Biazoli Ph.D. - Large-Scale Bibliometric Analysis of International Researcher Mobility in Brazilian Health and Life Sciences - Published in Quantitative Science Studies (2026) - DOI: 10.1162/QSS.a.411	Federal University of Lavras (UFLA) — Brazil 2022 - 2025
Clayton Ramos da Silva M.S. - Leishmaniasis Transmission Modeling Using Agent-Based Models and Complex Networks - Published in Revista Brasileira de Ensino de Física (2025) - DOI: 10.1590/1806-9126-RBEF-2024-0464	Federal University of Lavras (UFLA) — Brazil 2022 - 2024
Matheus de Andrade Flausino M.S. - Urban Topology and Street Crime Patterns Using Complex Networks - Published in Journal of Complex Networks (2025) - DOI: 10.1093/comnet/cnaf016	Federal University of Lavras (UFLA) — Brazil 2022 - 2024
Thiago Guedes de Jesus M.S. - Epidemiological Modeling with Evolutionary Game Theory in Spatial Networks	Federal University of Lavras (UFLA) — Brazil 2022 - 2024
Rolf Pagotto Veiga M.S. - Diretrizes para Classificação Automática de Músicas Cristãs: Aplicação do BERT na Identificação de Adoração M.S. defended August 30, 2024 — PPGCC/UFLA	Federal University of Lavras (UFLA) — Brazil 2021 - 2024
Alessandra Louzada Terra M.S. - Aprendizagem de Máquina para Estudo de Dados em Grafos: Um Estudo do Movimento dos Estudantes de Medicina no Brasil M.S. defended January 30, 2025 — PPGCC/UFLA	Federal University of Lavras (UFLA) — Brazil 2024 - 2025
Kelly Iapueque Rodrigues de Sousa M.S. - Urban Afforestation and Public Safety: A Bibliometric Study - Published in Revista de Gestão Ambiental e Sustentabilidade (GeAS) (2022) - DOI: 10.5585/geas.v11i2.22965	Federal University of Lavras (UFLA) — Brazil 2021 - 2023
Thiago do Prado Ramos M.S. - Wireless Network Capacity Planning from Campus Connection Logs	Federal University of Lavras (UFLA) — Brazil 2021 - 2023
Eduardo Vinicius Betim B.S. - Análise do Comportamento de Comunidade sobre Dados da Guerra da Ucrânia no Reddit	Universidade Federal de Santa Catarina (UFSC) — Brazil 2024 - 2024
Adilson Henrique de Oliveira Lima B.S. - Modelagem baseada em agentes para estudos sobre transmissão de HIV e comportamento voltado para a prevenção da disseminação do vírus Project interrupted (2023)	Federal University of Lavras (UFLA) — Brazil 2022 - 2023
Débora Rossini Martins Cardoso B.S. - Impactos do Ensino Remoto durante a Pandemia no Ensino Superior Brasileiro	Federal University of Lavras (UFLA) — Brazil 2022 - 2022

Eduardo Miranda Pedrosa Filho B.S. - Estudo e Implementação de Melhorias em um Sistema de Streaming para Vídeos de Educação Cristã	Federal University of Lavras (UFLA) — Brazil 2021 - 2021
Erich Fernando da Costa B.S. - Planejamento Policial a partir do Uso de Ferramentas de Business Intelligence: Análise do Índice de Criminalidade Violenta nos Municípios do 8º BPM	Federal University of Lavras (UFLA) — Brazil 2020 - 2020
Héuller Soares Vilela Silva B.S. - Previsão do Resultado Final de Partidas de Futebol a partir do Primeiro Tempo usando Técnicas de Aprendizagem de Máquina	Federal University of Lavras (UFLA) — Brazil 2020 - 2020
Ana Carolina Le Senechal B.S. - Análise e Pré-processamento de Dados Utilizando Técnicas de Mineração de Dados para o Moodle	Federal University of Lavras (UFLA) — Brazil 2013 - 2013
Giulia Luan Santos de Sousa B.S. - Tweetmining: Análise de Opinião Contida em Textos Extraídos do Twitter	Federal University of Lavras (UFLA) — Brazil 2012 - 2012
Larissa Avelino Zuppo B.S. - Webometria e Análise de Impacto da Informação em Redes Sociais	Federal University of Lavras (UFLA) — Brazil 2012 - 2012
Paulo Roberto de Almeida B.S. - Validação do Modelo Matemático para a Análise do Comportamento do Protocolo TCP em Redes Assimétricas	Federal University of Lavras (UFLA) — Brazil 2011 - 2011
Karen Aline Alves Bezerra B.S. - BDD: Avaliação e Proposta de Aplicação no Pós Desenvolvimento	Federal University of Lavras (UFLA) — Brazil 2011 - 2011
Arthur Moreira Maia B.S. - Modelagem da Estrutura e Dinâmicas de Redes Sociais: Um Modelo Formal para Estudos de Fenômenos Sociodinâmicos • Undergraduate research (Iniciação Científica) — CNPq	Federal University of Lavras (UFLA) — Brazil 2020 - 2020
Aline Rodrigues Guimarães de Oliveira B.S. - Modelagem e Simulação de Difusão de Informação em Redes Sociais • Undergraduate research (Iniciação Científica) — CNPq	Federal University of Lavras (UFLA) — Brazil 2019 - 2019
Álvaro Martins Espíndola B.S. - Uso de Modelagem Baseada em Agentes para Redução de Oportunidade para Agentes Criminosos na Cidade de Lavras • Undergraduate research (Iniciação Científica)	Federal University of Lavras (UFLA) — Brazil 2019 - 2019
Leonardo Amorim de Sena B.S. - Desenvolvimento e Análise de Plataforma para Estudos sobre Contágio em Redes Sociais • Undergraduate research (Iniciação Científica)	Federal University of Lavras (UFLA) — Brazil 2019 - 2019
Arthur Moreira Maia B.S. - Modelagem e Simulação de Difusão de Informação em Redes Sociais • Undergraduate research (Iniciação Científica) — CNPq	Federal University of Lavras (UFLA) — Brazil 2019 - 2019

Danielle Aparecida Alcântara B.S. - Webometria e Análise de Valor na Web • Undergraduate research (Iniciação Científica)	Federal University of Lavras (UFLA) — Brazil 2013 - 2013
Ana Carolina Le Senechal B.S. - Validação do Modelo Matemático para a Análise do Comportamento do Protocolo TCP em Redes Assimétricas • Undergraduate research (Iniciação Científica)	Federal University of Lavras (UFLA) — Brazil 2011 - 2011
Yasmin Ribeiro Penha B.S. - Estudo de Algoritmos para Mineração de Opinião para Detecção das Percepções sobre a Cidade de Lavras • Undergraduate research (Iniciação Científica)	Federal University of Lavras (UFLA) — Brazil 2011 - 2011
Ian Hawthorne, Nathan Holtrop B.S. - Teaching Ancient Wisdom with Modern AI: Evaluating LLMs in Early Christian Studies • McGregor Undergraduate Research Fellowship	Calvin University Summer 2026
Caleb Ausema B.S. - Cognitive Modeling of Churchgoers • Senior Project	Calvin University Spring 2026
Martin Liu B.S. - VerseNet: A Visual Neo4j-Based Bible Knowledge Network • Senior Project	Calvin University Spring 2026

University Service

Promotions Committee, Computer Science, Calvin University Responsible for the department's Promotions Committee: publicizing the Computer Science program and shaping recruitment strategy.	2026 – present
Planning team, School of Engineering and Computer Science, Calvin University Member of the team shaping Calvin's new School of Engineering and Computer Science.	2026 – present
Faculty planning group, CCCS grant on political polarization, Calvin University Member of a faculty group planning a Calvin Center for Christian Scholarship (CCCS) grant on political polarization.	2025 – present
Interviewer, prospective Honors students, Calvin University Interviews prospective students for the Calvin Honors Program.	2025 – present
Faculty Advisor, Girls Who Code, Calvin University Faculty advisor to the Girls Who Code chapter at Calvin University, including leading the annual student delegation to the Grace Hopper Celebration.	2025 – present

Professional Service

Faculty Fellow, Nagel Institute for the Study of World Christianity	Dec 2025
External Member - Masters Exam — Stephano Daniel Santos, UFLA Brazil	Aug 2025
External Member - PhD Qualifying Exam — Jamile Santos dos Santos, UFRJ Brazil	Aug 2024
External Member - Qualifying Exam — Rolf Pagotto Veiga, UFLA Brazil	Aug 2024

Church and Community Service

Elder - Plymouth Heights Christian Reformed Church, Grand Rapids, MI

May 2025 – present

Assists with liturgy; hospitality ministry to college students.

Professional Development

CCCS Writing Co-Op

Calvin University

Week-long writing residency: "Temperance and the Digital Life" (article for The Banner).

July 2026

G&M Young Leaders Training Conference

The Grace & Mercy Founda-

Participated in the Grace & Mercy Foundation Young Leaders Training Conference in New York (July 7–12, 2026), on implementing the Public Reading of Scripture and the Just Show Up book club at Calvin University. Trip funded by the Grace & Mercy Foundation and Calvin Theological Seminary.

tion — New York, NY

July 2026

Faith & Learning Fellowship

Calvin University
Sept 2025 – Dec 2025

Healthy Dialogue Series — Jennifer Powell McNutt

Calvin University

Faculty session with Jennifer Powell McNutt (February 18, 2025).

Feb 2025

CCCS Writing Co-Op

Calvin University

Week-long writing residency: "Prudence and the Digital Life."

Jan 2025

Healthy Dialogue Series — Felipe do Vale

Calvin University

Faculty/staff book discussion with author Felipe do Vale (November 6, 2024).

Nov 2024

Kuiper Seminar

Calvin University

Year-long seminar on faith and learning integration, pedagogy, and the Reformed tradition.

Sept 2024 – May 2025

Research Skills

Agent-based modeling: Netlogo and Python tools for ABMs.

Data Science and Machine Learning: Python, R, Statistics.

Social Network Analysis: Gephi, NetworkX.

Network-oriented modeling: Techniques for modeling complex systems using network-oriented methods.

Programming: Proficient with Python, C++, Netlogo, and Git; good understanding of Web development.

Languages: English (fluent), Portuguese (native), Dutch (basic).